# ANDISA KHUMALO

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## PROFESSIONAL PROFILE

Final-year Electrical & Computer Engineering student, South32 bursary holder and 3rd-best degree finalist, based locally and inclined towards heavy industry. Hands-on experience in PLC control, industrial automation and data analysis, including a PLC-based clean-room control system designed to strict safety standards, designed an ML-driven autonomous vehicle, and Python and MATLAB for analysing system performance. Detail-oriented, analytical and a strong team player with clear written and verbal communication. Seeking a Graduate Engineer role to support reliability, continuous improvement and operational excellence in industry.

## PROFESSIONAL ACTIVITIES & ACHIEVEMENTS

*2024 – Present*

**South African Institute of Electrical Engineers (SAIEE)**

*Student Member*

- Attended technical talks and industry workshops, deepening knowledge of electrical engineering practices and emerging industry trends.
- Networked with guest speakers, including professional engineers and mentors, building a professional network and gaining practical career insights.

*2026*

**SPETM Conference**

*Presenter*

- Presented a research poster on the design and implementation of an electric and autonomous vehicle at the international SPETM Conference (University of Johannesburg), attended by researchers from around the world.
- Engaged with international researchers and industry professionals, explaining the vehicle's system architecture, control strategy, and machine-learning workflow while answering technical questions.

## EDUCATION

*2023-Nov 2026*

**University of Zululand**

*Bachelor of Engineering, Electrical and Computer Engineering*

- Selected study areas: engineering systems design, Process control & Instrumentation, Control systems, software engineering, assembly language and operating systems, industrial engineering law, occupational health and safety, power systems, signals and systems and communications & networks.

*2021*

**Qhamuka High School**

*Secondary School (Matric)*

## PRACTICAL INVOLVEMENTS & CERTIFICATIONS

1. **South32 Bursar**
   - Bursary awarded on academic merit
2. **Eskom Espo program**
   - Evaluated approximately 10 engineering, electronics, robotics, coding and science projects, assessing technical functionality and project presentations against the programme's criteria.
   - Assessed projects, questioned participants on their technical work, provided feedback and contributed to the grading and selection process.
   - Presented assessment results with the programme team; one of the candidates assessed achieved 2nd position.

3. **OpenEuler certification (Linux systems administration), Huawei ICT Academy**
Certified in Linux system administration using OpenEuler, covering core Linux operating system management, configuration, and administration.

4. **OpenGauss certification (database administration), Huawei ICT Academy**
Certified in OpenGauss database administration, covering database management, configuration, and fundamental administration practices.

5. **SMR Machinery and Equipment (Distinction)**, National Research Nuclear University MEPhI (Russia) and University of Zululand, October 2026. Special seminar on small modular reactor machinery and equipment, completed as part of a joint training programme.

## TECHNICAL SKILLS

**Programming:** C#, Python, C/C++, MATLAB, Arduino programming, Kotlin/Android development
**Industrial Automation:** Siemens PLC programming, ladder logic, SCADA concepts, PID control, Instrumentation
**Software & Web:** ASP.NET Core MVC, HTML/CSS, SQL server, EF Core, GIT/GITHUB **Embedded Systems:** Raspberry Pi, Arduino, STM32, serial communication, sensor & camera integration, motor control
**Computer Vision & AI:** OpenCV, lane keeping, machine learning workflows, camera data collection & model deployment
**Control & Simulation:** MATLAB/Simulink, PID design, modelling and engineering simulation
**Engineering Documentation:** IEEE-format technical reports, system architecture documentation, database administration

## ENGINEERING & TECHNOLOGY

*2026*

### Autonomous Electric Vehicle - Final-Year Capstone

- Designed, built and tested a low-cost autonomous RC vehicle powered by a Raspberry Pi 4B, integrating camera-based perception, embedded control and motor actuation.
- Developed and trained a convolutional neural network (CNN) for lane-following steering using 1,620 labelled image–steering pairs, achieving a steering-angle MAE of 16.66° and RMSE of 25.03°.
- Integrated an OV5647 camera, PCA9685 PWM controller, steering servo, BTS7960 motor driver and ultrasonic sensors for autonomous steering and experimental obstacle detection.
- Developed an Android application using Kotlin and Jetpack Compose, with a Flask backend on the Raspberry Pi to support live video streaming, manual vehicle control and autonomous-mode initialisation.

*2026*

**AK Mechanics - Full-Stack Vehicle Fleet Management System**

- Built a full-stack web application using ASP.NET Core MVC, Entity Framework Core and SQL Server.
- Implemented role-based access control for clients, service providers and administrators using ASP.NET Identity.
- Deployed the application using Microsoft Azure App Service and Azure SQL Database.

*2026*

**Smart Airport Luggage Tracking System - Capstone Group Project**

- Developed the proposed tracking architecture covering RFID identification, GPS positioning, IoT sensing, communication between luggage and the central system, database integration and web/app interfaces.
- Designed the proposed tracking and communication approach around RFID, LoRa and MQTT, including the tracking algorithm and sensor subsystem. • Contributed to the system proposal and execution plan; project outputs were presented as a proposed solution rather than a deployed system.

## Additional Information

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**Driver's Licen**ce: Code 10
**Location:** Richards Bay, Ready to start work after November 2026

## REFERENCES

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**Dr OE Oni from University of Zululand**

• onio@unizulu.ac.za

**Mr. B Khoza from University of Zululand**

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